

Notes: Atkeson and Burstein (JPE, 2019)

Aggregate Implications of Innovation Policy

Contents

1	Big picture	3
2	Data and measurement (key objects)	4
2.1	Aggregate data sources and calibration inputs	4
2.2	Firms, products, and innovation categories	4
2.3	Key aggregate objects	5
2.4	Balanced growth path and timing	5
3	Models and empirical specifications (all equations + interpretation)	5
3.1	Environment with innovation by entrants only	5
3.2	Innovation intensity and research labor	6
3.3	Analytical results with entry only: productivity, output, welfare, and fiscal cost . . .	7
3.4	Full model with incumbent innovation	9
3.5	Impact elasticities in the full model and conditional efficiency	9
4	Identification and instruments (what makes the estimates credible)	10
4.1	What standard approaches miss	10
4.2	What identifies the sufficient statistics	10
4.3	How V is measured	11
4.4	Main calibrated sufficient statistics	11
4.5	Why the estimates are credible	11
4.6	Relation to the literature	12
5	Tables	12
5.1	Table 1: Elasticities of productivity, output, welfare, and optimal innovation intensity	12
6	Short summary	13

7	Conclusion	14
7.1	Core conclusion (what the paper establishes)	14
7.2	Economic intuition	14
7.3	Takeaway	14

1 Big picture

- **Question.** What quantitative impact should policy-induced changes in firms' innovative investment have on aggregate productivity and output at different horizons, what are the welfare implications, and what is the fiscal cost of such policies?
- **Setting.** The paper studies a growth model with firm innovation that nests several canonical frameworks in the literature. It starts with a simple model in which only **entering firms** innovate, then extends it to a fuller model in which **incumbents** also innovate by **acquiring new products** and **improving existing products**.
- **Core challenge.** Different innovation-growth models imply very different transition dynamics after policy changes. The paper wants a way to compare them in a unified quantitative language instead of solving each model from scratch and comparing model-specific objects.
- **Main methodological contribution.** The paper shows that, up to a first-order approximation, the response of aggregate productivity and output to innovation policy can be summarized by **two sufficient statistics**:
 - the **impact elasticity of aggregate productivity growth** with respect to aggregate real innovative investment, denoted V ;
 - the **degree of intertemporal knowledge spillovers in research**, denoted f .
- **Why this matters.** Once V and f are measured, the model's positive implications for productivity and output dynamics, and many of its normative implications for welfare and optimal policy, become transparent.
- **Main quantitative message.** Uniform innovation subsidies of reasonable magnitude can have only **modest effects on aggregate productivity and output over a 20-year horizon**, even though the **welfare gains can still be substantial** because innovation policy affects very long-run productivity.
- **Main measurement result.**
 - In a specification with **no business stealing**, the paper measures $V \approx 0.026$.
 - In a specification **with business stealing**, it measures $V \approx 0.010$.
 - For intertemporal knowledge spillovers, the paper considers a **low-spillover case** $f = -1.6$ and a **high-spillover case** $f = 0.96$.
- **Main substantive results.**
 - A policy that raises the allocation of labor to research enough to move innovation intensity from $\bar{i}_r = 0.089$ to $\bar{i}'_r = 0.10$ has a long-run fiscal cost of about **1.1% of output per year**.
 - With low spillovers, long-run productivity and output responses are modest.
 - With high spillovers, long-run responses become extremely large, but medium-run effects remain much smaller.
 - Welfare gains can be large because households value the long-run productivity effects a lot.
- **Economic takeaway.** The key bottleneck is not simply “how much R&D spending rises,” but **how strongly innovative investment translates into productivity growth on impact**

and **how much current research lowers the cost of future research**. Medium-run policy effects are limited unless these sufficient statistics are favorable.

2 Data and measurement (key objects)

2.1 Aggregate data sources and calibration inputs

- **National accounts data.** The paper uses NIPA data for the **US nonfinancial corporate sector** to measure:
 - growth of output,
 - the physical-capital share a ,
 - the markup m ,
 - dividends relative to output,
 - incumbent innovative investment relative to output.
- **Firm dynamics data.** The paper uses the **Longitudinal Business Database** to measure the employment share of new firms, which helps pin down the contribution of entrants when the model assumes no business stealing.
- **External growth-accounting inputs.**
 - Akcigit and Kerr (2018) and Garcia-Macia, Hsieh, and Klenow (2016) provide estimates of entrants' contribution to productivity growth.
 - Jones (2002), Fernald and Jones (2014), and Bloom et al. (2017) provide estimates or guidance for the knowledge-spillover parameter f .
- **Unit of time.** One period in the quantitative exercise is **one year**.

2.2 Firms, products, and innovation categories

- **Consumption-good side.** A final consumption good is produced from a **continuum of differentiated intermediate goods** aggregated with a CES technology.
- **Research-good side.** A separate **research good** is produced with research labor and is the direct input into innovation.
- **Entering firms.** Entrants invest in the research good to acquire one product. Their innovation may either:
 - create a **new product for society**, or
 - **steal** an existing product from an incumbent while improving its productivity.
- **Incumbent firms.** In the full model, incumbents also innovate in two distinct ways:
 - **acquiring new products** new to the firm,
 - **improving continuing products** the firm already owns.

2.3 Key aggregate objects

- **Aggregate productivity:** Z_t .
- **Measure of available products:** M_t .
- **Research-good output:** Y_{rt} .
- **Innovation intensity of the economy:**

$$i_{rt} \equiv \frac{P_{rt}Y_{rt}}{Y_t}.$$

- **Research labor share:**

$$l_{rt} \equiv \frac{L_{rt}}{L_t}, \quad l_{pt} \equiv \frac{L_{pt}}{L_t}.$$

- **Sufficient statistics:**

- V : impact elasticity of productivity growth with respect to aggregate real innovative investment,
- f : intertemporal knowledge spillovers in research.

2.4 Balanced growth path and timing

- **Balanced growth path (BGP).** On the BGP, output and capital grow at constant rates, aggregate productivity grows at rate $\bar{g}_Z$, innovation intensity is constant, and real research output is constant.
- **Semiendogenous vs. endogenous growth.**
 - If $f < 1$, the model is **semiendogenous**: long-run growth depends on exogenous scientific progress and population growth.
 - If $f = 1$ and $\bar{g}_A = \bar{g}_L = 0$, the model becomes **endogenous**: long-run growth is driven directly by firms' innovation decisions.
- **Policy experiment.** The main quantitative experiment is a **permanent proportional change in innovation subsidies** that raises the allocation of labor to research.

3 Models and empirical specifications (all equations + interpretation)

3.1 Environment with innovation by entrants only

The final consumption good satisfies the resource constraint

$$Y_t = C_t + K_{t+1} - (1 - d_k)K_t. \quad (1)$$

A representative household has preferences

$$\sum_{t=0}^{\infty} \beta^t \frac{L_t (C_t/L_t)^{1-\gamma}}{1-\gamma}. \quad (2)$$

Intermediate good z is produced with

$$y_t(z) = z k_t(z)^a l_t(z)^{1-a}. \quad (3)$$

The final good is a CES aggregate,

$$Y_t = \left(\sum_z y_t(z)^{\frac{r-1}{r}} M_t(z) \right)^{\frac{r}{r-1}}, \quad (4)$$

which implies the aggregate production representation

$$Y_t = Z_t K_t^a L_{pt}^{1-a}, \quad (5)$$

with aggregate productivity

$$Z_t = \left(\sum_z z^{r-1} M_t(z) \right)^{\frac{1}{r-1}}. \quad (6)$$

The research good is produced as

$$Y_{rt} = A_{rt} Z_t^{f-1} L_{rt}. \quad (8)$$

The total measure of products evolves as

$$M_{t+1} = [1 - d_0 + (1 - d_e)x_{et}] M_t, \quad (9)$$

and aggregate productivity growth satisfies

$$g_t^Z \equiv \log Z_{t+1} - \log Z_t = G(Y_{rt}), \quad (10)$$

where

$$G(Y_r) = \frac{1}{r-1} \log(1 - d_0 + (h_e - d_e)Y_r). \quad (11)$$

Interpretation.

- The consumption-good block is a standard CES differentiated-products environment.
- Aggregate productivity Z_t rises either because there are **more products** or because product quality/productivity improves.
- The research-good technology is where the parameter f enters: if $f < 1$, higher Z_t makes future research harder, so knowledge spillovers are incomplete.
- In the entrant-only model, all innovation is summarized by real research investment Y_{rt} and its effect on productivity through the function $G(\cdot)$.

3.2 Innovation intensity and research labor

The price of the research good equals marginal cost,

$$P_{rt} = \frac{Z_t^{1-f}}{A_{rt}} W_t. \quad (12)$$

Innovation intensity is

$$i_{rt} = \frac{P_{rt}Y_{rt}}{Y_t},$$

and labor allocated to research satisfies

$$l_{rt} = \frac{i_{rt}}{i_{rt} + (1-a)/m}, \quad l_{pt} = 1 - l_{rt}. \quad (13)$$

Interpretation.

- The policy object the paper emphasizes is not raw R&D spending alone, but the **innovation intensity of the economy**, that is, innovative expenditure relative to output.
- A higher innovation intensity reallocates labor away from current production and toward research.
- This creates a short-run output cost, even before any productivity gain materializes.

3.3 Analytical results with entry only: productivity, output, welfare, and fiscal cost

Long-run productivity. If policy permanently changes the innovation intensity and therefore the research labor share, then in the semiendogenous case the long-run productivity gap across BGPs is

$$\log Z'_t - \log \bar{Z}_t = \frac{1}{1-f} (\log \bar{l}'_r - \log \bar{l}_r). \quad (14)$$

Interpretation.

- Long-run productivity depends only on the change in research labor and on f .
- If f is far below 1, the long-run elasticity is small.
- If f is near 1, long-run effects become huge because the model approaches endogenous growth.

Impact elasticity. The paper defines

$$V \equiv \frac{dG(Y_r)}{dY_r} \bar{Y}_r, \quad (15)$$

and interprets V as the **impact elasticity of aggregate productivity growth** with respect to aggregate real innovative investment.

To a first-order approximation,

$$\log Z'_{t+1} - \log Z'_t - \bar{g}_Z \approx V (\log Y'_{rt} - \log \bar{Y}_r). \quad (16)$$

Transition dynamics can then be written as

$$\log Z'_{t+1} - \log \bar{Z}_{t+1} \approx \sum_{j=0}^t G_j (\log l'_{r,t-j} - \log \bar{l}_r), \quad (17)$$

with

$$G_0 = V, \quad G_{j+1} = [1 - (1-f)V]G_j. \quad (18)$$

Interpretation.

- V controls the **short-run impact** of more research on productivity growth.
- f controls how much of that effect persists through time.
- So V matters most for the medium run, while f is decisive for the long run.

Output dynamics. The path of output is

$$\log Y'_t - \log \bar{Y}_t = \frac{1}{1-a} (\log Z'_t - \log \bar{Z}_t) + (\log l'_{pt} - \log \bar{l}_p) - \frac{a}{1-a} (\log R'_{kt} - \log \bar{R}_k). \quad (20)$$

Across BGPs, long-run output satisfies

$$\log \bar{Y}'_t - \log \bar{Y}_t = \left[\frac{1}{(1-a)(1-f)} - \frac{\bar{l}_r}{\bar{l}_p} \right] (\log \bar{l}'_r - \log \bar{l}_r). \quad (21)$$

Interpretation.

- Output falls on impact because labor is reallocated from production to research.
- Over time, output rises because productivity accumulates.
- Whether policy looks powerful at a 20-year horizon depends on whether the cumulative productivity gain is large enough to offset the initial labor reallocation cost.

Welfare and the socially optimal research share. The consumption-equivalent welfare change is approximated by

$$\log y \approx (1 - \tilde{\beta}) \sum_{t=0}^{\infty} \tilde{\beta}^t \frac{\bar{Y}_t}{\bar{C}_t} \left[(\log Z'_t - \log \bar{Z}_t) + (1-a)(\log l'_{pt} - \log \bar{l}_p) \right]. \quad (22)$$

The socially optimal BGP allocation of labor to research is

$$\frac{\bar{l}_r^*}{\bar{l}_p^*} = \frac{1}{1-a} \frac{\tilde{\beta}V}{1 - \tilde{\beta}[1 - (1-f)V]}. \quad (24)$$

Interpretation.

- Welfare trades off a **short-run consumption/output loss** against a **long-run productivity gain**.
- The socially optimal research share is increasing in V and in persistence coming from f .

Fiscal cost. For a permanent change in subsidies,

$$\log \bar{l}'_r - \log \bar{l}_r = \log(1 - \bar{t}_e) - \log(1 - \bar{t}'_e),$$

and

$$\frac{\bar{E}'}{\bar{Y}'} - \frac{\bar{E}}{\bar{Y}} = \bar{l}'_r - \bar{l}_r. \quad (\text{Proposition 4})$$

Interpretation.

- In the long run, the increase in innovation intensity equals the increase in subsidy expenditure as a share of output.
- This gives the paper a very transparent fiscal interpretation of policy experiments.

Upper bound on the impact elasticity. In the entrant-only model,

$$V \leq \bar{g}_Z - G(0). \quad (26)$$

Interpretation.

- The impact elasticity cannot exceed the contribution of innovative investment to trend productivity growth.
- This is just a concavity result: the marginal product of innovative investment is bounded by its average product.

3.4 Full model with incumbent innovation

In the full model, innovative investment has three components:

$$x_{ct} + x_{mt} + x_{et} = Y_{rt}. \quad (27)$$

Here:

- x_{et} is entrant innovation,
- x_{mt} is incumbent investment to **acquire new products**,
- x_{ct} is incumbent investment to **improve continuing products**.

Aggregate productivity growth becomes

$$G(x_c, x_m, x_e) = \frac{1}{r-1} \log \left((1 - d_0 - d_m h(x_m) - d_e x_e) z(x_c) + h_m h(x_m) + h_e x_e \right). \quad (28)$$

Interpretation.

- The full model is rich enough to nest both **expanding-varieties models** and **neo-Schumpeterian quality-ladder models**.
- Entry can create new goods or steal old ones; incumbents can either improve what they already have or acquire additional goods.
- Yet the aggregate law of motion remains tractable because the model is built so innovative investment scales with product size.

3.5 Impact elasticities in the full model and conditional efficiency

The paper defines category-specific impact elasticities

$$V_i = \frac{\partial G}{\partial x_i} \bar{Y}_r, \quad i \in \{c, m, e\}. \quad (29)$$

Any policy-induced overall elasticity is then

$$V = V_c \frac{dx_c}{dY_r} + V_m \frac{dx_m}{dY_r} + V_e \frac{dx_e}{dY_r}. \quad (30)$$

If the allocation of innovative investment on the initial BGP is **conditionally efficient**, then

$$V_c = V_m = V_e,$$

and the same reduced-form analytical results from the entrant-only model continue to apply.

Interpretation.

- The sufficient-statistics result survives in the full model if the baseline composition of innovative investment is efficient enough.
- If not, then the effect of innovation policy depends not only on how much total R&D rises, but also on how R&D is reallocated across entry, product acquisition, and improvement of continuing goods.

Under a **proportional policy change**, however, the allocation of real innovative investment on the new BGP is the same as on the initial BGP, and the earlier formulas remain usable with an appropriately measured V .

4 Identification and instruments (what makes the estimates credible)

4.1 What standard approaches miss

- A direct model-by-model approach makes quantitative comparisons difficult because different innovation models have different states, transitions, and policy margins.
- Standard growth-accounting approaches that treat innovative investment like ordinary intangible capital often miss the fact that social and private returns can differ because of:
 - love of variety,
 - business stealing,
 - intertemporal spillovers in research productivity.

4.2 What identifies the sufficient statistics

- **The spillover parameter f .** The paper does not estimate f internally; instead it studies quantitative implications under two externally motivated values:
 - **low spillovers:** $f = -1.6$,
 - **high spillovers:** $f = 0.96$.
- **The impact elasticity V .** The paper measures an upper bound on V by measuring V_e , the impact elasticity associated with entrant innovation.

4.3 How V is measured

- In the **no-business-stealing** case,

$$V = V_e = \frac{1}{r-1} \frac{\bar{Y}_r}{\bar{x}_e} \bar{s}_e, \quad (38)$$

where $\bar{s}_e$ is the size share of entering firms. This is measured using the employment share of new firms in the Longitudinal Business Database.

- In the **business-stealing** case, the contribution of entrants to productivity growth is not equal to entrant size because some entrant products displace incumbents. The paper therefore uses external estimates that entrants account for about **25.7% of annual trend productivity growth**.
- The share of aggregate innovative investment undertaken by entrants, $\bar{x}_e/\bar{Y}_r$, is not directly observed in NIPA because entrants may invest before incorporation. The paper infers it from:
 - the zero-profit entry condition,
 - the value of incumbent firms,
 - dividends and intangible capital measures.
- The resulting measurement gives

$$\frac{\bar{x}_e}{\bar{Y}_r} \approx 0.34.$$

4.4 Main calibrated sufficient statistics

- The elasticity of substitution is set to $r = 4$.
- With **no business stealing**, the paper obtains

$$V \approx 0.026.$$

- With **business stealing**, it obtains

$$V \approx 0.010.$$

- These numbers are small enough that medium-run policy effects on aggregate output are limited.

4.5 Why the estimates are credible

- The key bound on V comes from concavity and is therefore economically transparent.
- The model is designed so that the same sufficient-statistics logic applies across a large class of innovation models, not just one special case.
- The paper is explicit that results are sensitive to the measurement of entrants' innovation share and to assumptions about business stealing, so it presents multiple calibrations rather than pretending there is a single exact answer.

TABLE 1
ELASTICITY OF AGGREGATE PRODUCTIVITY, OUTPUT, AND WELFARE WITH RESPECT TO
RESEARCH LABOR AT VARIOUS TIME HORIZONS, AND OPTIMAL INNOVATION INTENSITY

INTERTEMPORAL SPILLOVERS	NO BUSINESS STEALING ($\delta_r = 0$)		WITH BUSINESS STEALING ($\delta_r = .2$)	
	Low ϕ (-1.6)	High ϕ (.96)	Low ϕ (-1.6)	High ϕ (.96)
Aggregate productivity:				
1. Year 1, Θ	.026	.026	.010	.010
2. Year 20	.29	.52	.16	.20
3. Long run	.38	25.00	.38	25.00
Aggregate output:				
4. Year 0	-.11	-.11	-.11	-.11
5. Year 20	.20	.42	.03	.07
6. Long run	.36	33.18	.36	33.18
7. Welfare (equivalent variation)	.26	2.01	.17	.73
8. Optimal share of labor in research	.30	.69	NA	NA

NOTE.—NA: not applicable.

4.6 Relation to the literature

- The paper nests and compares **Jones-style semiendogenous growth**, **quality-ladder Schumpeterian growth**, and **expanding-varieties growth**.
- It also connects to the literature on **innovation as intangible investment**, but emphasizes that innovation policies cannot be evaluated with standard growth-accounting formulas alone when externalities and business stealing matter.

5 Tables

5.1 Table 1: Elasticities of productivity, output, welfare, and optimal innovation intensity

Read:

- Table 1 is the paper's main quantitative summary. It reports elasticities of aggregate productivity, output, and welfare with respect to the allocation of labor to research under four calibrations:
 - no business stealing + low f ,
 - no business stealing + high f ,
 - business stealing + low f ,
 - business stealing + high f .
- **Impact elasticities of productivity.** The year-1 elasticity is exactly V :
 - 0.026 with no business stealing,
 - 0.010 with business stealing.

This is the immediate productivity-growth response to more innovative investment.

- **Twenty-year productivity effects.** By year 20, productivity elasticities are still moderate:
 - 0.29 and 0.52 in the no-business-stealing model,
 - 0.16 and 0.20 in the business-stealing model.
- **Long-run productivity effects.** These are entirely driven by f :

- with low spillovers, the long-run elasticity is only 0.38;
- with high spillovers, it jumps to 25.00.
- **Impact effect on output.** Output falls on impact in all four cases, with elasticity -0.11 . This is the direct cost of moving labor from production into research.
- **Twenty-year output effects.** By year 20 the elasticity of output is only:
 - 0.20 or 0.42 with no business stealing,
 - 0.03 or 0.07 with business stealing.

The paper’s main medium-run message comes from these numbers: **uniform innovation subsidies do not move aggregate output very much over 20 years.**

- **Long-run output effects.** These are modest with low f (0.36) but huge with high f (33.18), again showing that very long-run results are dominated by the knowledge-spillover assumption.
- **Welfare.** Welfare elasticities are positive in all cases:
 - 0.26 and 2.01 with no business stealing,
 - 0.17 and 0.73 with business stealing.

So even when medium-run output effects are small, welfare gains can be meaningful because future productivity gains matter a lot in present value.

- **Optimal research labor.** In the conditionally efficient no-business-stealing case, the socially optimal share of labor in research is much larger than the baseline level:
 - 0.30 under low spillovers,
 - 0.69 under high spillovers.

Main interpretation.

- The table shows very clearly that V **matters for medium-run effects** and f **matters for long-run effects**.
- It also explains the paper’s headline result: **reasonable innovation subsidies may have modest effects on 20-year productivity and output, but much larger welfare effects.**

6 Short summary

- **Method.** The paper develops a unified sufficient-statistics approach to innovation policy in a model that nests several canonical growth-through-innovation frameworks.
- **Identification.** The key objects are the impact elasticity V and the research spillover parameter f . The paper measures V using data on entrants’ contribution to growth and entrants’ share of innovative investment, and studies alternative calibrated values of f .
- **Application.** Using aggregate US data and external measurements, the paper calibrates four model variants that differ in whether business stealing is present and whether intertemporal spillovers are weak or strong.

- **Main result.** Uniform innovation subsidies of plausible magnitude have only **modest medium-run effects** on aggregate productivity and output, but can still generate **substantial welfare gains** because of long-run growth effects.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper shows that the aggregate effects of innovation policy can be summarized using two sufficient statistics, V and f , across a broad class of models.
- It also shows that the quantitative effect of uniform subsidies to innovative investment is likely to be **small over a 20-year horizon** unless those sufficient statistics are especially favorable.
- At the same time, the welfare gains from such subsidies can still be large because households value the cumulated long-run productivity gains.

7.2 Economic intuition

- Innovation policy works through two margins:
 - an **impact margin**: how much extra current innovative investment raises productivity growth now (V),
 - a **persistence margin**: how much today’s research affects the productivity of future research (f).
- Raising research effort has an immediate cost because labor leaves current production.
- The policy looks successful only if the induced productivity gains cumulate quickly enough and persist strongly enough.
- This is why the paper finds a weak link between innovation subsidies and medium-run aggregate output even when welfare effects remain positive.

7.3 Takeaway

This paper is valuable because it turns a large and messy innovation-policy literature into a clean quantitative map. Instead of asking which exact canonical innovation model is “right,” it asks which **sufficient statistics** matter for policy and how large they are. The answer is sobering for medium-run policy evaluation: reasonable innovation subsidies are unlikely to produce dramatic aggregate output gains over 20 years. But the answer is not pessimistic overall, because long-run productivity effects can still make such policies welfare improving.